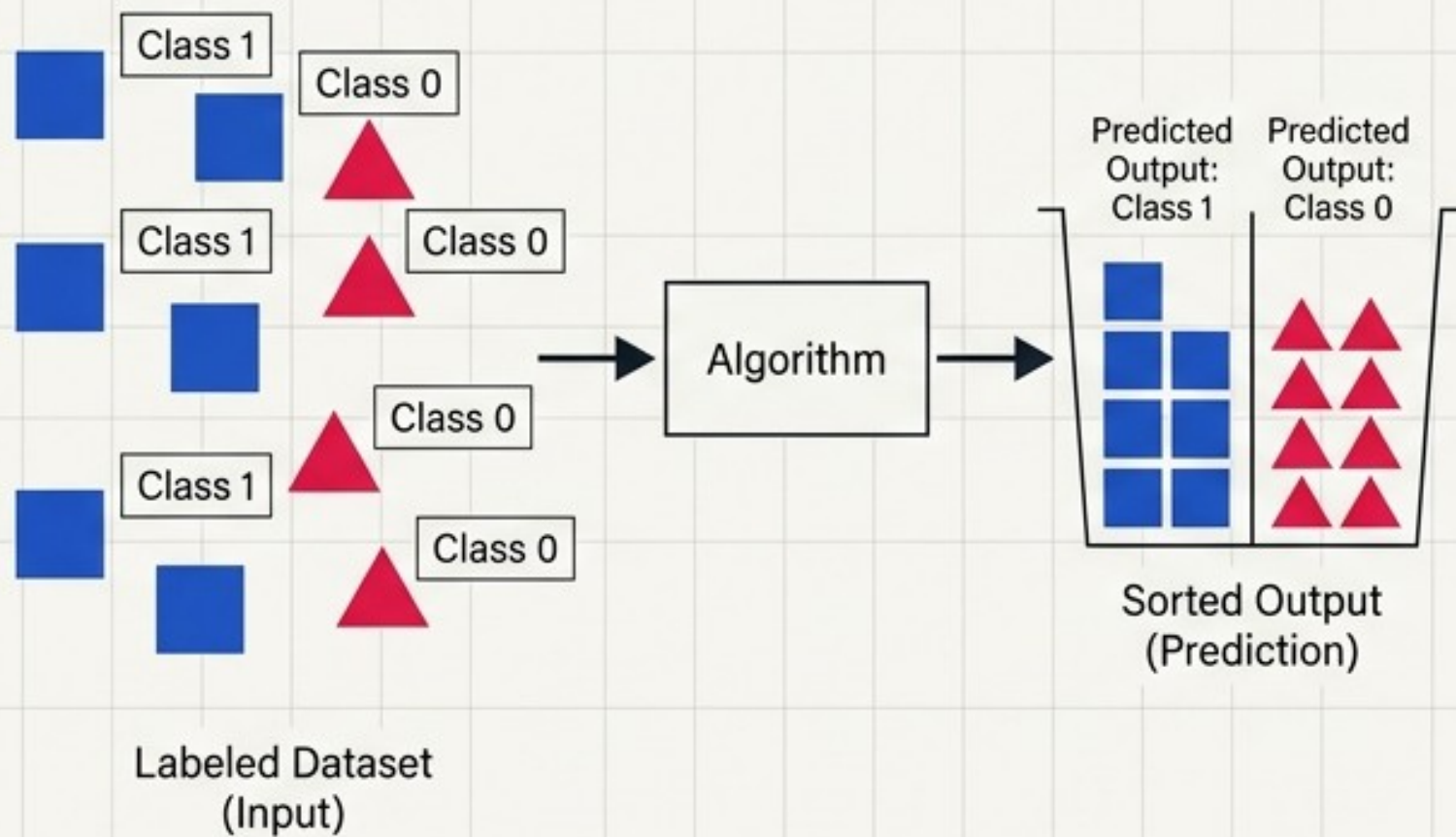
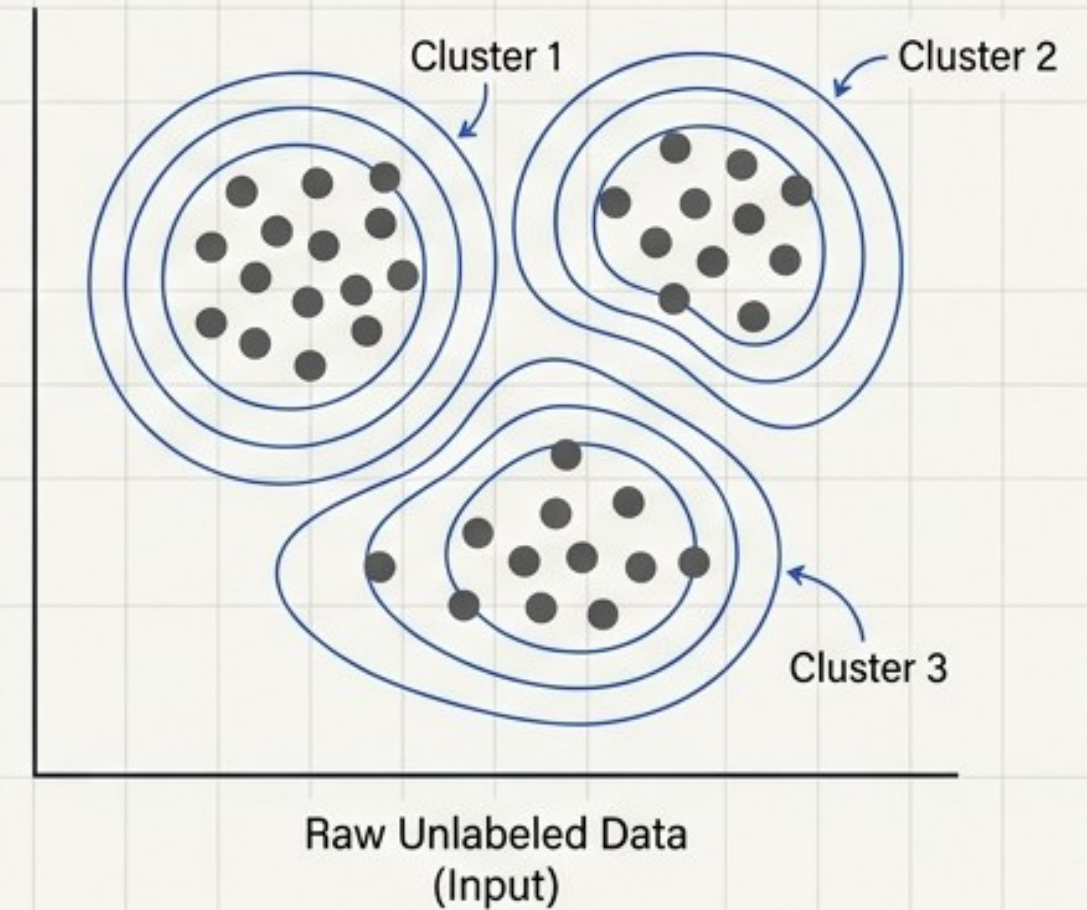


Machine learning divides broadly into learning from labeled answers versus discovering hidden structures.

Supervised Learning



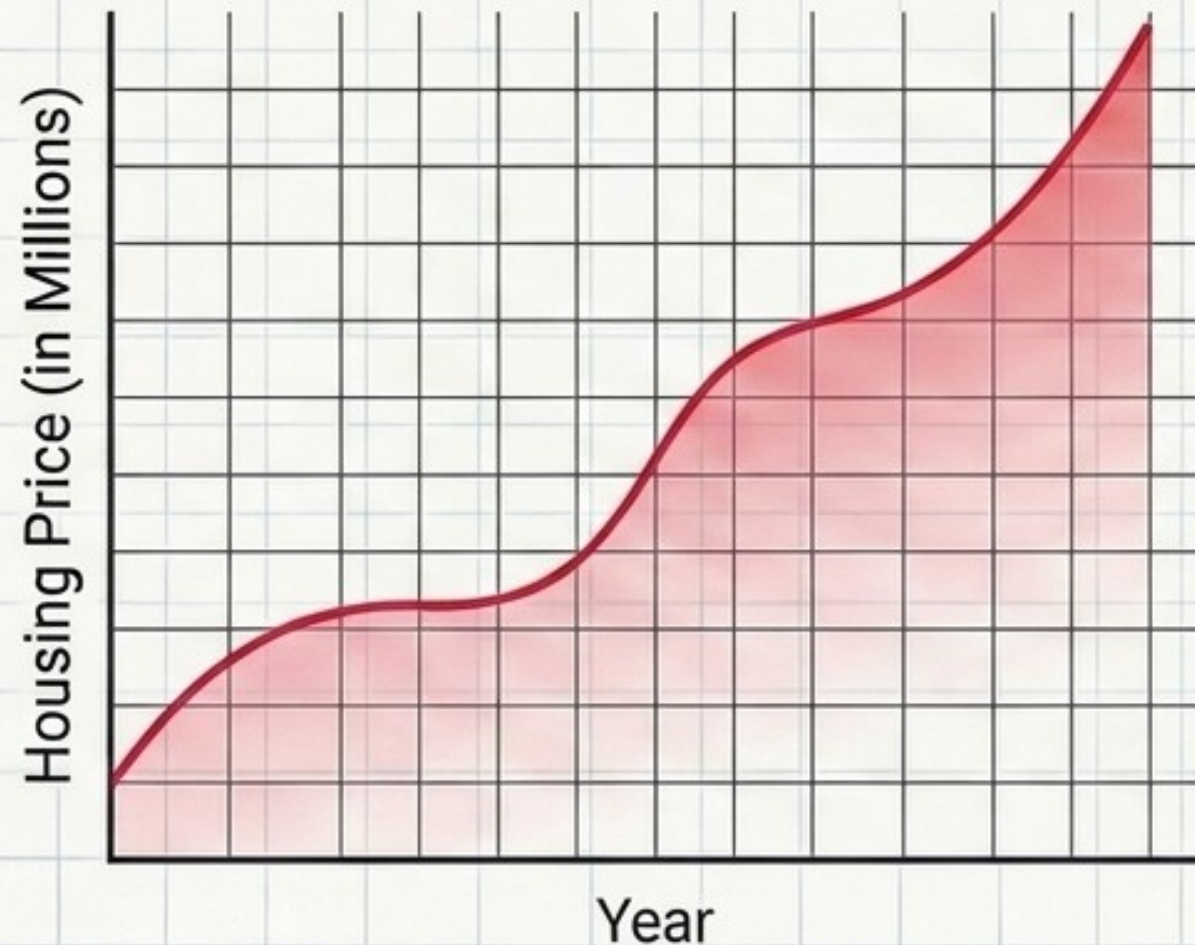
Unsupervised Learning



- **Supervised Learning:** Trains on labeled historical data to map inputs to known outputs.
- **Unsupervised Learning:** Explores raw, unlabeled data to discover inherent patterns or clusters.

Supervised learning asks two fundamentally different questions: "How much?" versus "Which one?"

Regression (Continuous)



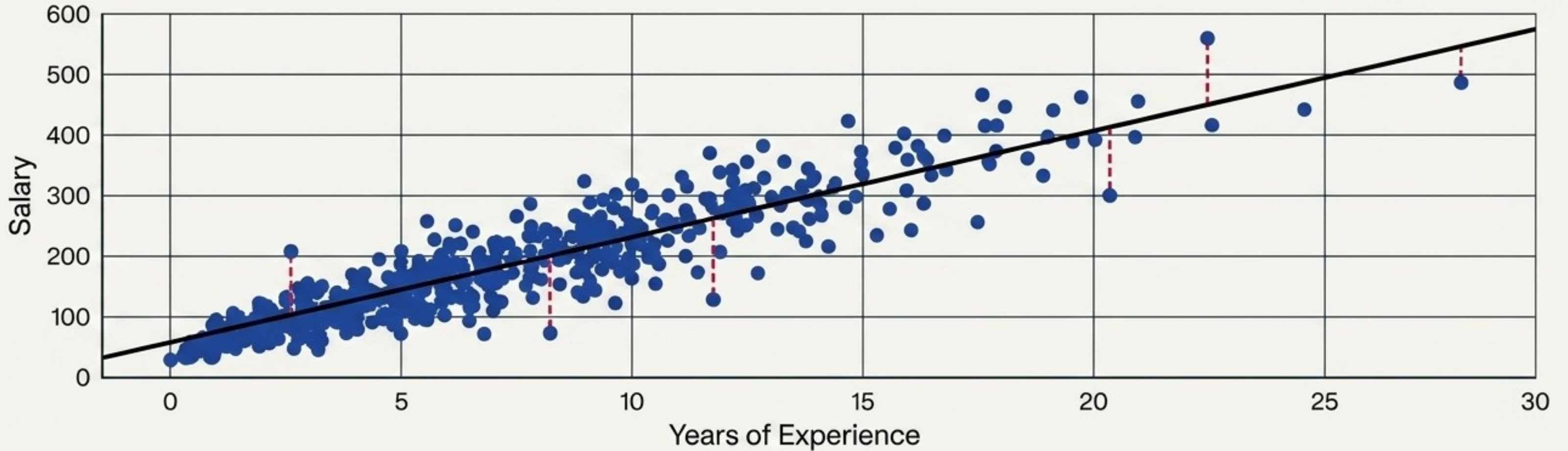
Predicts unbounded, continuous numeric values (e.g., pricing, temperature).

Classification (Discrete)



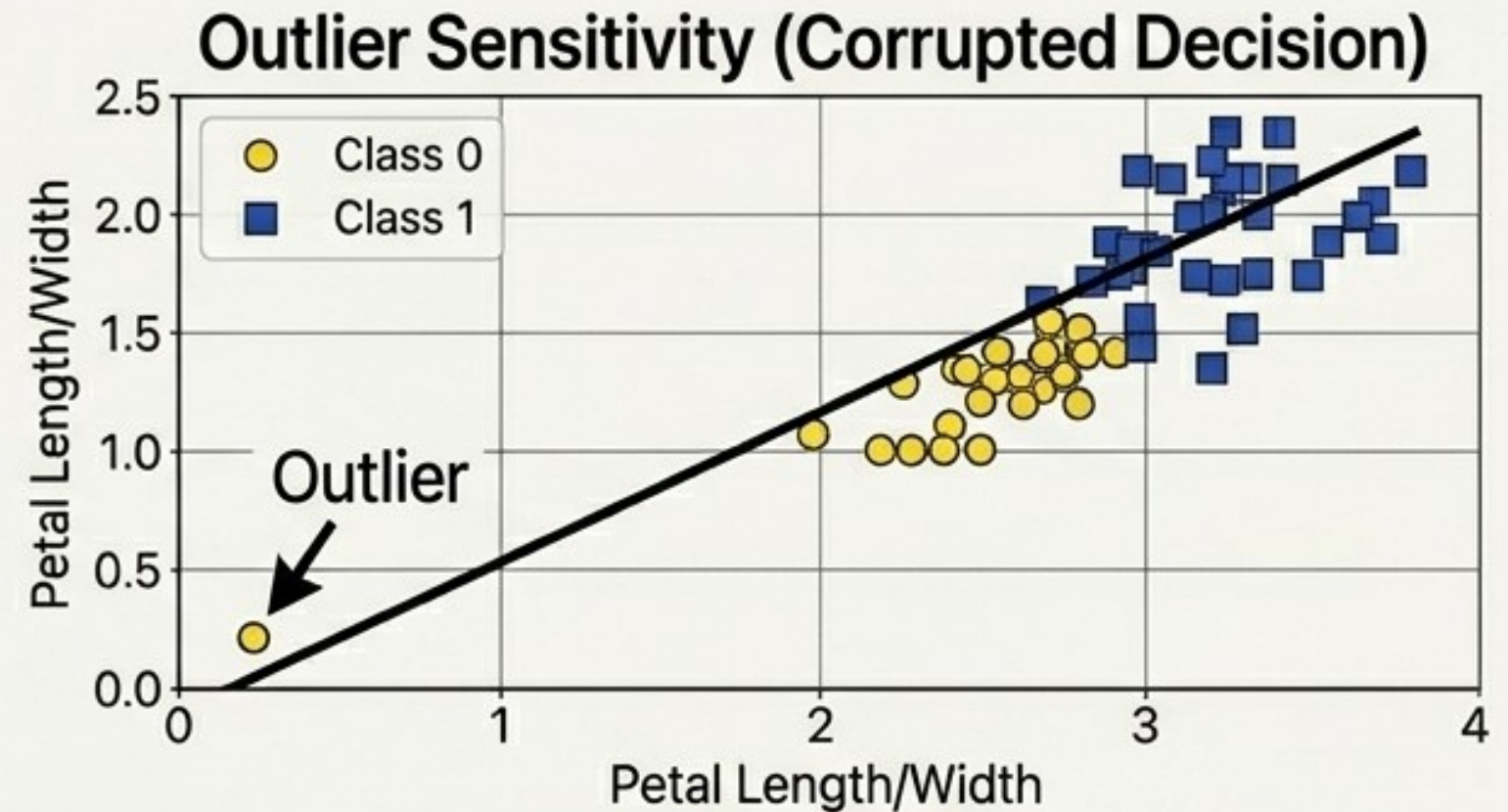
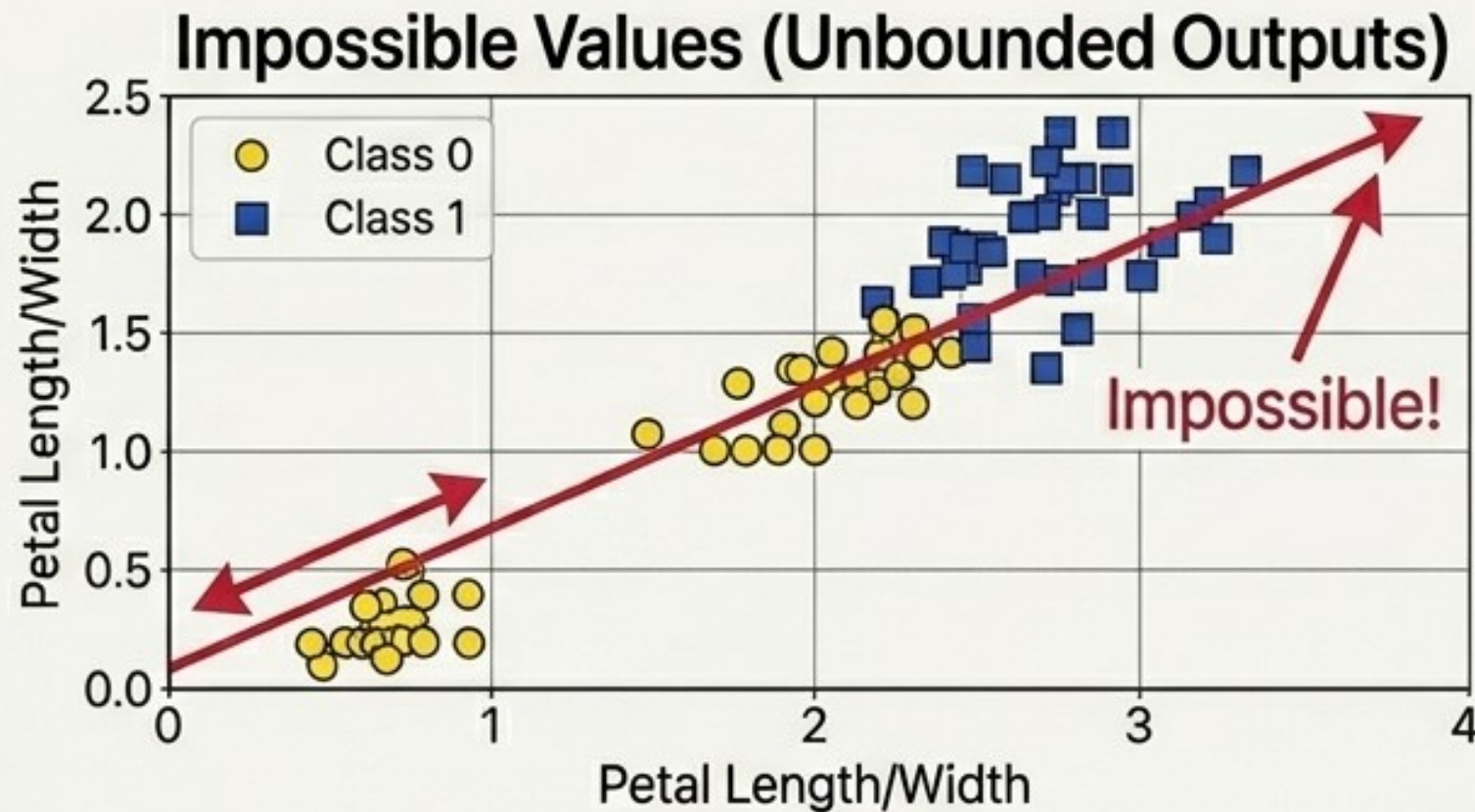
Predicts discrete, categorical class labels based on a target vector of separate groups.

Linear regression finds the best-fitting straight line to predict continuous outputs.



- Calculates a weighted sum of input features to draw a continuous mathematical trendline.
- Optimizes the model by minimizing the overall distance between predictions and actual data points.
- Functions flawlessly for modeling relationships where variables scale infinitely.

Linear regression fails for classification because it predicts impossible values and breaks on outliers.

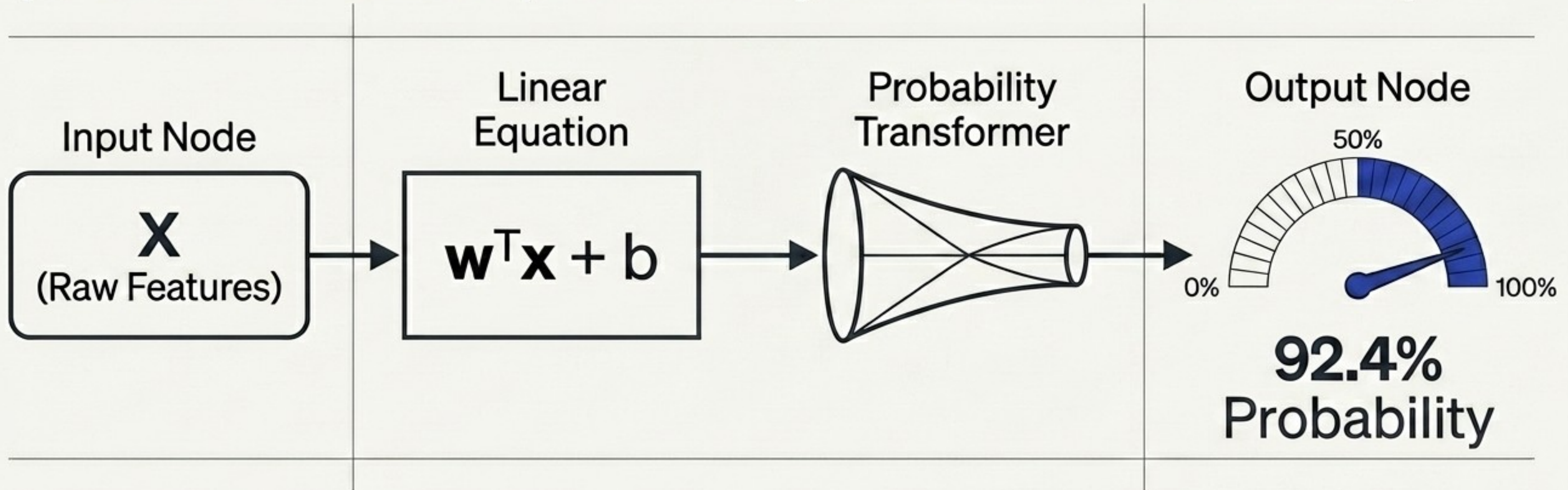


Mathematical Flaws:

- **Unbounded Outputs:** Straight lines inevitably predict probabilities lower than 0% or greater than 100%.
- **Outlier Sensitivity:** Extreme, isolated data points instantly corrupt the line's decision accuracy.

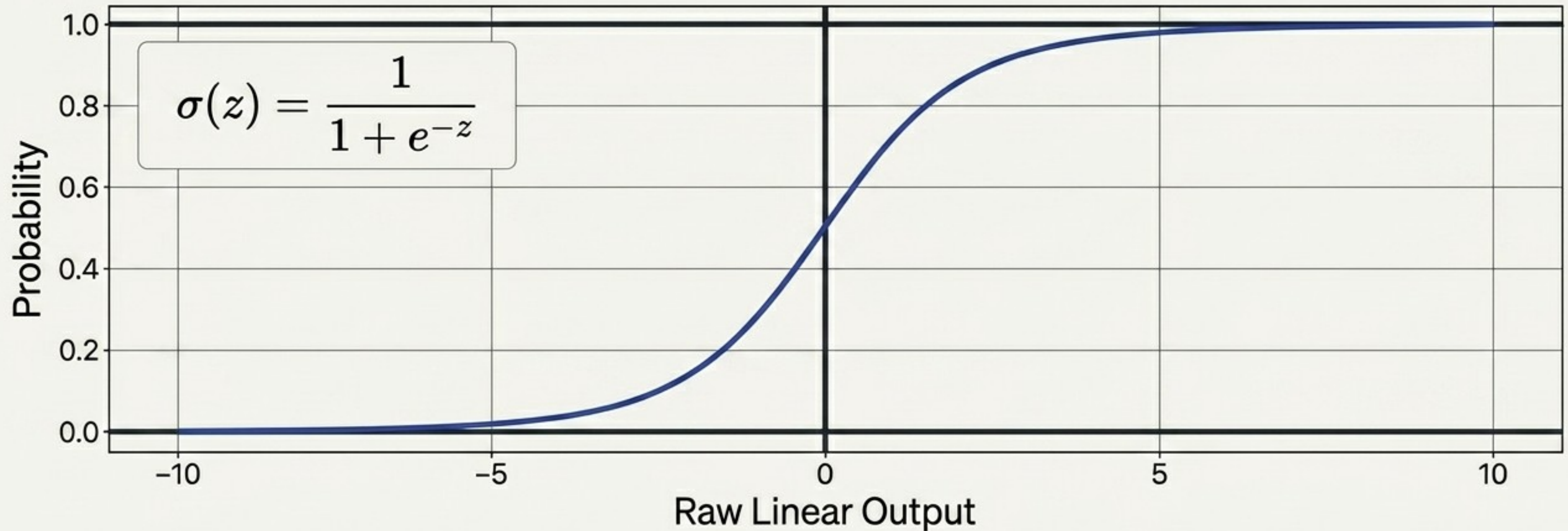
Linear Regression Formula: $y = \mathbf{w}^T \mathbf{x} + b$ (Vector Notation) or $y = \theta_1 x_1 + \theta_2 x_2 + \dots + b$ (Scalar Notation)

Logistic regression adapts the linear model to predict the exact probability of class membership.



- Maintains the exact same linear combination of features used in standard regression.
- Passes the raw numeric output through a transformation block to yield a strict probability.
- Creates a mathematical bridge between continuous calculations and categorical decisions.

The Sigmoid function compresses any linear output into an S-shaped curve strictly bound between 0 and 1.

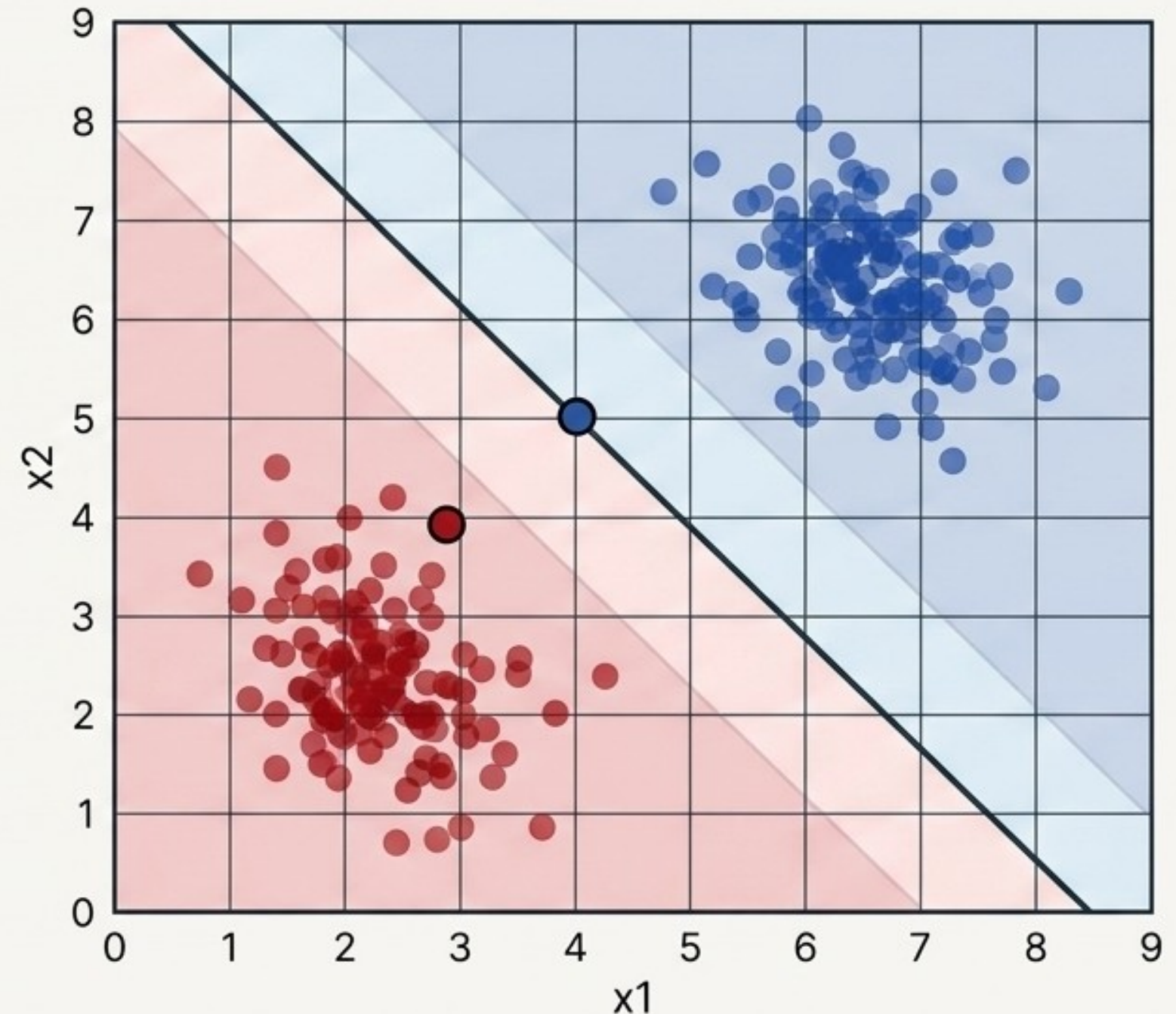


- **Asymptotic Bounds:** Extreme inputs mathematically approach, but never exceed, 0 or 1.
- **The Squashing Effect:** Forces all predictions into a valid, interpretable probability distribution.

The model separates classes by drawing a decision boundary where probability is equally split.

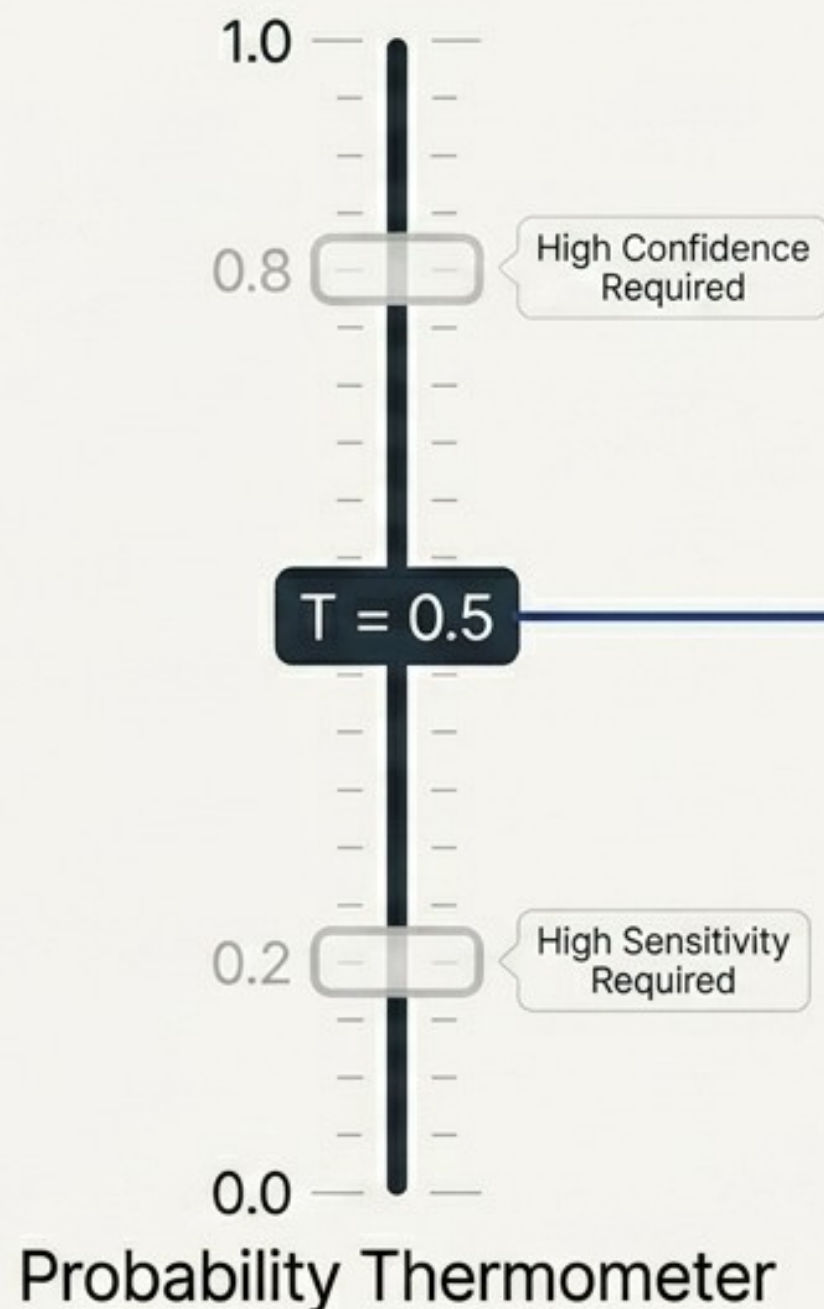
Spatial Separation

- Maps multi-dimensional feature space into distinct, color-coded classification zones.
- The boundary represents spatial coordinates where model confidence is exactly split at 50%.
- Calculates the margins to cleanly isolate discrete historical observations.



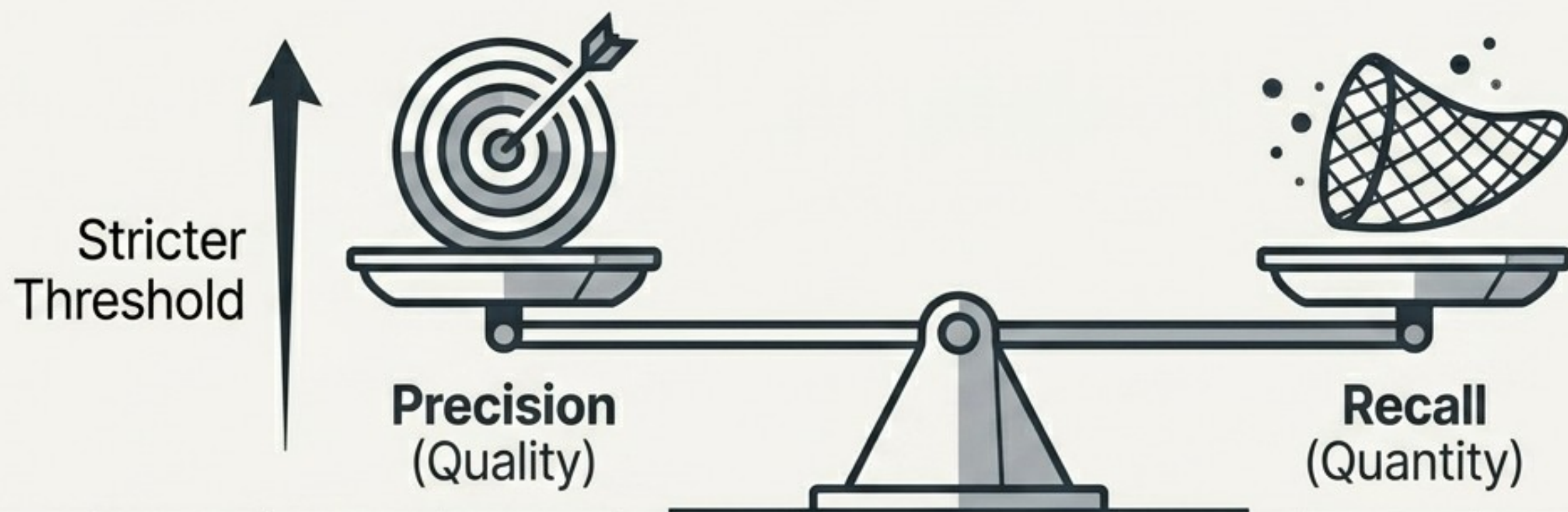
Converting continuous probabilities into discrete decisions requires defining a classification threshold.

- **Default Baseline ($T=0.5$):**
The standard mathematical cutoff for deciding between two binary classes.



- **Strategic Adjustments:**
Thresholds map math to real-world consequences, tailored to business risk tolerance.
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Thresholds map math to real-world consequences, tailored to business risk tolerance.

Shifting the classification threshold creates an inescapable tradeoff between Precision and Recall.



Precision:

Out of all positive predictions the model made, how many were actually correct?

Recall:

Out of all the actual positive cases in reality, how many did the model successfully find?

The Inherent Tradeoff:

Optimizing for exactness almost always degrades total discovery, and vice versa.

Linear regression forecasts exact quantities, while logistic regression classifies via probability



Linear Regression

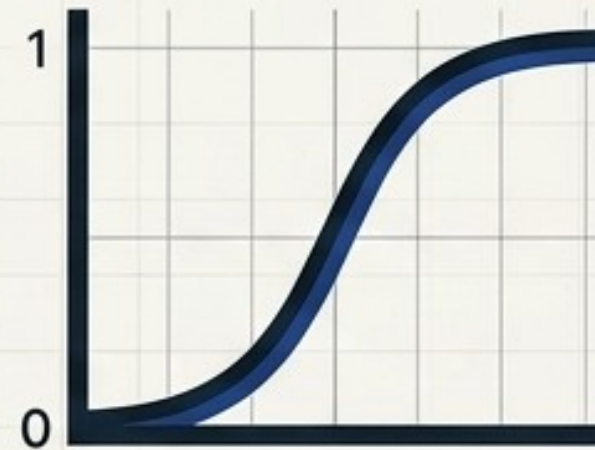
Output:

Unbounded Continuous Number

Primary Use:

Forecasting, Pricing, Trend Lines

- Deploy Linear Regression exclusively when the target variable is continuous and theoretically infinite.



Logistic Regression

Output:

Bounded Probability [0-1]

Primary Use:

Binary Classification, Risk Scoring, Spam Detection

- Deploy Logistic Regression when the target represents distinct, discrete categories.

• The Sigmoid function serves as the ultimate mathematical bridge, transforming linear equations into categorical logic.